

Incidence of the main genetic markers in glioblastoma multiforme is independent of tumor topology.

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Glioblastoma multiforme (GBM) is the most common as well as the most aggressive type of primary brain tumor of astrocytic origin in adults. GBM is characterized by a high degree of intratumoral heterogeneity both in histomorphology and genetic changes. Trisomy/polysomy of chromosome 7, monosomy of chromosome 10, EGFR gene amplification and p53 deletion have been described as the typical genetic markers for tumor classification and prediction of possible response to therapy. Our work was based on detection of these four main genetic changes both in central and peripheral parts of the tumors to evaluate possible differences in the topological incidence of these genetic markers. Chromosomal abnormalities in tumor samples from a group of 21 patients surgically treated for GBM were characterized by means of the interphase-fluorescence in situ hybridization (I-FISH) technique using sets of centromere and locus-specific DNA probes. In addition, we performed a detailed analysis of one selected tumor sample using a genomic microarray system (GenoSensor Array 300) to characterize copy number changes of specific sequences and refine results obtained by I-FISH. However, the data show no significant differences in occurrence of the described genetic markers in either part of the tumor.

Chromosome 7 Gain Compensates for Chromosome 10 Loss in Glioma.

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The co-occurrence of chromosome 10 loss and chromosome 7 gain in gliomas is the most frequent loss-gain co-aneuploidy pair in human cancers. This phenomenon has been investigated since the late 1980s without resolution. Expanding beyond previous gene-centric studies, we investigated the co-occurrence in a genome-wide manner, taking an evolutionary perspective. Mining of large-scale tumor aneuploidy data confirmed the previous finding of a small-scale longitudinal study that the most likely order is chromosome 10 loss, followed by chromosome 7 gain. Extensive analysis of genomic and transcriptomic data from both patients and cell lines revealed that this co-occurrence can be explained by functional rescue interactions that are highly enriched on chromosome 7, which could potentially compensate for any detrimental consequences arising from the loss of chromosome 10. Transcriptomic data from various normal, noncancerous human brain tissues were analyzed to assess which tissues may be most predisposed to tolerate compensation of chromosome 10 loss by chromosome 7 gain. The analysis indicated that the preexisting transcriptomic states in the cortex and frontal cortex, where gliomas arise, are more favorable than other brain regions for compensation by rescuer genes that are active on chromosome 7. Collectively, these findings suggest that the phenomenon of chromosome 10 loss and chromosome 7 gain in gliomas is orchestrated by a complex interaction of many genes residing within these two chromosomes and provide a plausible reason why this co-occurrence happens preferentially in cancers originating in certain regions of the brain. Significance: Increased expression of multiple rescuer genes on the gained chromosome 7 could compensate for the downregulation of several vulnerable genes on the lost chromosome 10, resolving the long-standing mystery of this frequent co-occurrence in gliomas.

Drak/STK17A Drives Neoplastic Glial Proliferation through Modulation of MRLC Signaling.

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Glioblastoma (GBM) and lower grade gliomas (LGG) are the most common primary malignant brain tumors and are resistant to current therapies. Genomic analyses reveal that signature genetic lesions in GBM and LGG include copy gain and amplification of chromosome 7, amplification, mutation, and overexpression of receptor tyrosine kinases (RTK) such as EGFR, and activating mutations in components of the PI3K pathway. In *Drosophila melanogaster*, constitutive co-activation of RTK and PI3K signaling in glial progenitor cells recapitulates key features of human gliomas. Here we use this *Drosophila* glioma model to identify death-associated protein kinase (Drak), a cytoplasmic serine/threonine kinase orthologous to the human kinase STK17A, as a downstream effector of EGFR and PI3K signaling pathways. Drak was necessary for glial neoplasia, but not for normal glial proliferation and development, and Drak cooperated with EGFR to promote glial cell transformation. Drak phosphorylated Sqh, the *Drosophila* ortholog of nonmuscle myosin regulatory light chain (MRLC), which was necessary for transformation. Moreover, Anillin, which is a binding partner of phosphorylated Sqh, was upregulated in a Drak-dependent manner in mitotic cells and colocalized with phosphorylated Sqh in neoplastic cells undergoing mitosis and cytokinesis, consistent with their known roles in nonmuscle myosin-dependent cytokinesis. These functional relationships were conserved in human GBM. Our results indicate that Drak/STK17A, its substrate Sqh/MRLC, and the effector Anillin/ANLN regulate mitosis and cytokinesis in gliomas. This pathway may provide a new therapeutic target for gliomas. **Significance:** These findings reveal new insights into differential regulation of cell proliferation in malignant brain tumors, which will have a broader impact on research regarding mechanisms of oncogene cooperation and dependencies in cancer. See related commentary by Lathia, p. 1036.

Arm-specific multicolor fluorescence in situ hybridization reveals widespread chromosomal instability in glioma cell lines.

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An investigation of numerical and structural chromosome aberrations using chromosome arm-specific multicolor fluorescence in situ hybridization (armFISH) revealed considerable genetic heterogeneity among and within 11 glioma cell lines. Despite the substantial variation in numerical chromosome alterations among the cell lines, several distinct and glioma growth-associated losses or gains were frequently observed, that is, losses of chromosomes 10, 13, and 22 and gain of chromosome 7 in particular. Structural aberrations frequently affected chromosomes 1, 4, 7, 16, and 19; however, no single structural chromosome aberration common to all or even several glioma cell lines could be found. Structural alterations were often multiform, and a large variety of unstable chromosome structures were detected. Two of the cell lines also harbored small marker chromosomes containing mainly heterochromatin and chromosomal insertions within heterochromatic regions. Altogether, the armFISH provides a versatile tool for the identification of chromosomal aberrations as well as their formation patterns in tumors with a complex genome at the level of chromosome arms.

The consensus molecular subtypes of colorectal cancer.

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Colorectal cancer (CRC) is a frequently lethal disease with heterogeneous outcomes and drug responses.

To resolve inconsistencies among the reported gene expression-based CRC classifications and facilitate clinical translation, we formed an international consortium dedicated to large-scale data sharing and analytics across expert groups. We show marked interconnectivity between six independent classification systems coalescing into four consensus molecular subtypes (CMSs) with distinguishing features: CMS1 (microsatellite instability immune, 14%), hypermutated, microsatellite unstable and strong immune activation; CMS2 (canonical, 37%), epithelial, marked WNT and MYC signaling activation; CMS3 (metabolic, 13%), epithelial and evident metabolic dysregulation; and CMS4 (mesenchymal, 23%), prominent transforming growth factor- β activation, stromal invasion and angiogenesis. Samples with mixed features (13%) possibly represent a transition phenotype or intratumoral heterogeneity. We consider the CMS groups the most robust classification system currently available for CRC-with clear biological interpretability-and the basis for future clinical stratification and subtype-based targeted interventions.

Molecular pathological classification of colorectal cancer.

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Colorectal cancer (CRC) shows variable underlying molecular changes with two major mechanisms of genetic instability: chromosomal instability and microsatellite instability. This review aims to delineate the different pathways of colorectal carcinogenesis and provide an overview of the most recent advances in molecular pathological classification systems for colorectal cancer. Two molecular pathological classification systems for CRC have recently been proposed. Integrated molecular analysis by The Cancer Genome Atlas project is based on a wide-ranging genomic and transcriptomic characterisation study of CRC using array-based and sequencing technologies. This approach classified CRC into two major groups consistent with previous classification systems: (1) ~16 % hypermutated cancers with either microsatellite instability (MSI) due to defective mismatch repair (~13 %) or ultramutated cancers with DNA polymerase epsilon proofreading mutations (~3 %); and (2) ~84 % non-hypermutated, microsatellite stable (MSS) cancers with a high frequency of DNA somatic copy number alterations, which showed common mutations in APC, TP53, KRAS, SMAD4, and PIK3CA. The recent Consensus Molecular Subtypes (CMS) Consortium analysing CRC expression profiling data from multiple studies described four CMS groups: almost all hypermutated MSI cancers fell into the first category CMS1 (MSI-immune, 14 %) with the remaining MSS cancers subcategorised into three groups of CMS2 (canonical, 37 %), CMS3 (metabolic, 13 %) and CMS4 (mesenchymal, 23 %), with a residual unclassified group (mixed features, 13 %). Although further research is required to validate these two systems, they may be useful for clinical trial designs and future post-surgical adjuvant treatment decisions, particularly for tumours with aggressive features or predicted responsiveness to immune checkpoint blockade.

Gene expression profiles in genome instability-based classes of colorectal cancer.

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Broad copy number aberrations (BCNAs) represent a common form of genome instability in colorectal cancer (CRC). CRCs show large variations in their level of aneuploidy: microsatellite-unstable (MSI) tumors are known to have a near-diploid karyotype while microsatellite-stable (MSS) tumors show high level of chromosomal instability. However, MSS tumors have great heterogeneity in the number of BCNAs, with a minor percentage of samples showing an almost normal karyotype. In the present work we subdivided

MSS CRCs according to a "BCNA score" and characterized their transcriptome profiles, considered as a proxy to their phenotypic features. Microsatellite testing, genome-wide DNA copy number and whole-transcript expression analysis (HTA) were performed on 33 tumor samples and 25 normal colonic tissue samples from 32 CRC patients. 15.1% of the samples were MSI tumors ($n = 5$), whereas 84.9% were MSS tumors ($n = 28$). Gene expression data of 34 additional MSI tumors was retrieved from a public functional genomics data repository. Using as a threshold the first quartile of the BCNA score distribution, MSS samples were classified as low-BCNA (LB, $n = 7$) or high-BCNA (HB, $n = 21$). LB tumors were enriched for mucinous CRCs and their gene-expression profile resembled that of MSI samples for what concerns a subset of genes involved in secretory processes, mucosal protection, and extracellular matrix remodeling. HB tumors were predominantly non-mucinous adenocarcinomas and showed overexpression of a subset of genes typical of surface colonocytes and EGF signaling. A large percentage of unclassified samples according to the consensus molecular subtypes (CMS) classifier was found in the LB group (43%), whereas 76% HB tumors belonged to CMS2. A classification of colorectal tumors based on the number of BCNAs identifies two groups of MSS tumors which differ for histopathology and gene expression profile. Such information can be exploited for its translational relevance in different aspects of CRC clinical management.

MiOXSYS® and OxiSperm® II assays appear to provide no clinical utility for determining oxidative stress in human sperm-results from repeated semen collections.

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Oxidative stress in semen contributes up to 80% of all infertility diagnosis. Diagnostics to measure oxidative stress in semen was recently added to the 6th edition WHO methods manual, although diagnostic predictive values need to be interpreted with caution as there are still several research questions yet to be answered. To determine the natural fluctuations in semen redox indicators (MiOXSYS® and OxiSperm® II) within and between men and their association with markers of sperm oxidative stress. Total, 118 repeat semen samples from 31 generally healthy men aged 18-45 years, over 6 months. Standard semen analysis as per 5th WHO manual. Semen redox levels measured via MiOXSYS® and OxiSperm® II. Additional attributes of sperm quality; HBA® binding assay and sperm hyperactivation and oxidative stress; DNA fragmentation (Halo® Sperm) and lipid peroxidation (BODIPY™ 581/591 C11) were assessed. Samples with high redox-potential (MiOXSYS® ≥ 1.47 sORP/106 sperm/ml) had lower sperm, motility, morphology and higher DNA fragmentation ($P < 0.05$). Upon further analysis, these associations were driven solely by the adjustment of sperm concentration (106 /ml) in normalised redox-potential. No significant associations between NBT-reactivity (OxiSperm® II) and measures of the sperm function or oxidative stress were observed ($P > 0.05$). Fluctuations in semen redox levels varied greater between men than within men over the study period. Neither MiOXSYS® nor OxiSperm® II assays were predictive of sperm function or sperm oxidative stress. This was likely due at least in part to limited understanding of their biochemistry and clinical application. As a result, these assays seem to provide no additional clinical utility beyond that of a standard semen analysis, highlighting the imperative for the development of new robust point-of-care devices for accurately determining sperm oxidative stress. These findings suggest that MiOXSYS® and OxiSperm® II systems for the measurement of sperm oxidative stress may have limited diagnostic potential.

Oxidation-reduction potential as a new marker for oxidative stress: Correlation to male

infertility.

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Male infertility affects men worldwide. Oxidative stress (OS), characterized by an overabundance of reactive oxygen species (ROS) or a deficiency of antioxidants, is one of the major causes of male infertility. OS causes damage at the molecular level, which impairs lipids, proteins, and DNA. The cyclic cascade of redox reactions weakens sperm function which leads to poor semen parameters and eventual sterility. There is a need for advanced diagnostic tests that can quickly and accurately detect OS. Most commonly used assays can only measure single constituents of OS. However, the MiOXSYS System introduces a new strategy to detect OS by measuring the oxidation-reduction potential (ORP)--a direct evaluation of the redox balance between ROS and antioxidants. The MiOXSYS System has shown promise as a diagnostic tool in the evaluation of male infertility. This review explores the concept of ORP, details the principle of the MiOXSYS System, and summarizes the findings in clinical studies that support ORP measurement in semen.

Redox Balance in Male Infertility: Excellence through Moderation-"Μέτρον ἄριστον".

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Male infertility, a relatively common and multifactorial medical condition, affects approximately 15% of couples globally. Based on WHO estimates, a staggering 190 million people struggle with this health condition, and male factor is the sole or contributing factor in roughly 20-50% of these cases. Nowadays, urologists are confronted with a wide spectrum of conditions ranging from the typical infertile male to more complex cases of either unexplained or idiopathic male infertility, requiring a specific patient-tailored diagnostic approach and management. Strikingly enough, no identifiable cause in routine workup can be found in 30% to 50% of infertile males. The medical term male oxidative stress infertility (MOSI) was recently coined to describe infertile men with abnormal sperm parameters and oxidative stress (OS), including those previously classified as having idiopathic infertility. OS is a critical component of male infertility, entailing an imbalance between reactive oxygen species (ROS) and antioxidants. ROS abundance has been implicated in sperm abnormalities, while the exact impact on fertilization and pregnancy has long been a subject of considerable debate. In an attempt to counteract the deleterious effects of OS, urologists resorted to antioxidant supplementation. Mounting evidence indicates that indiscriminate consumption of antioxidants has led in some cases to sperm cell damage through a reductive-stress-induced state. The "antioxidant paradox", one of the biggest andrological challenges, remains a lurking danger that needs to be carefully avoided and thoroughly investigated. For that reason, oxidation-reduction potential (ORP) emerged as a viable ancillary tool to basic semen analysis, measuring the overall balance between oxidants and antioxidants (reductants). A novel biomarker, the Male infertility Oxidative System (MiOXSYS®), is a paradigm shift towards that goal, offering a quantification of OS via a quick, reliable, and reproducible measurement of the ORP. Moderation or "Μέτρον" according to the ancient Greeks is the key to successfully safeguarding redox balance, with MiOXSYS® earnestly claiming its position as a guarantor of homeostasis in the intracellular redox milieu. In the present paper, we aim to offer a narrative summary of evidence relevant to redox regulation in male reproduction, analyze the impact of OS and reductive stress on sperm function, and shed light on the "antioxidant paradox" phenomenon. Finally, we examine the most up-to-date scientific literature regarding ORP and its measurement by the recently developed MiOXSYS® assay.